Chen 2020 NPC — recapitulation summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8/14 samples (≥107 cells) intermix on one continuous manifold: iLISI 2.1 / 3.2 ceiling, metrics plateau at θ≥2 (over-integration null passes); P11-removal control confirms genuine cross-patient mixing; single kNN component.

Gate 2 — module geography → NO-GO

No Gast/pSom hub. Gast and pSom are orthogonal (r = −0.05) — neither co-varying (overlap case) nor cross-repressing (branching); both diffuse across the manifold. Structure is carcinoma-state (cycling, squamous/basal, hypoxia, IFN/MHC-II), not embryonic. Both oppose Crest (−0.54/−0.44); Crest is not a proliferation confound.

Gate 3 — hub mutual exclusivity → NA

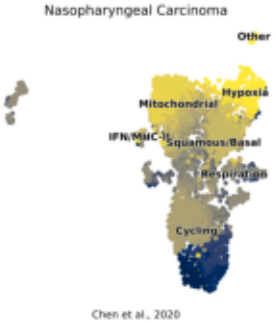
Not tested — no bifurcation and no hub to run 04_correlation/05_modules on (branching-only gate).

STEP FIGURES

Gate 1 — samples



Trajectory / root



Gast signature



pSom signature



Crest signature



EMT-TF expression



Dataset. Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 14 patients / 3,763 malignant load; 3,674 across 8 samples (≥107 cells) enter the manifold.

Verdict. ✗ Pattern does not reproduce. Gate 1 GO but Gate 2 NO-GO: NPC malignant cells form a continuous carcinoma-state manifold with no embryonic module geography — Gast/pSom independent, no hub, no arms. EMT-TFs PRRX1 (62%) / VIM (53%) broad, no mesenchymal terminus; canonical SNAI1/ZEB1/TWIST1 near-absent (1–7%).

Parameter choices

Min cells/sample	≥ 107 (keeps 8/14)	**Integration**	Harmony on sample, θ=2, 20 it
Island removal	none (single kNN comp.)	**Trajectory**	scFates PPT N=50 λ=300 σ=0.5
Root node	35 = Progenitor-like	**n_pcs**	19 (>0.5% var each; θ-sweep 0–10)
Scaling	none (log1p, reference-matched)	**Signatures**	score_genes, union control pool